All 15 Experiments - Flutter & PWA

EXP 1: Install and Configure Flutter Environment

Steps:

1. Go to https://flutter.dev/docs/get-started/install
2. Download Flutter SDK for your OS (Windows/Mac/Linux)
3. Extract the zip to `C:\flutter` (Windows) or `~/flutter` (Mac/Linux)
4. Add Flutter to PATH:
   - Windows: Edit Environment Variables → Add `C:\flutter\bin`
   - Mac/Linux: Add `export PATH="$PATH:~/flutter/bin"` to `.bashrc` or `.zshrc`
5. Install Android Studio from https://developer.android.com/studio
6. In Android Studio → SDK Manager → Install Android SDK
7. Run `flutter doctor` in terminal to check setup
8. Fix any issues shown by flutter doctor
9. Create first project: `flutter create my_app`
10. Run the app: `cd my_app` then `flutter run`

Commands:

```bash
flutter doctor
flutter create my_app
cd my_app
flutter run
```

EXP 2: Design Flutter UI Using Common Widgets

File: `lib/main.dart`

```dart
```

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Common Widgets Demo',
      theme: ThemeData(primarySwatch: Colors.blue),
      home: MyHomePage(),
    );
  }
}

class MyHomePage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Common Widgets'),
        backgroundColor: Colors.blue,
      ),
      body: SingleChildScrollView(
        padding: EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            // Text Widget
            Text(
              'Hello Flutter!',
              style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
            ),
            SizedBox(height: 10),

            // Button Widget
            ElevatedButton(
              onPressed: () {
                print('Button Pressed');
              },
              child: Text('Click Me'),
            ),
            SizedBox(height: 10),
```

```
// TextField Widget
TextField(
  decoration: InputDecoration(
    labelText: 'Enter your name',
    border: OutlineInputBorder(),
  ),
),
SizedBox(height: 10),

// Image Widget
Image.network(
  'https://flutter.dev/images/flutter-logo-sharing.png',
  height: 100,
),
SizedBox(height: 10),

// Container Widget
Container(
  width: double.infinity,
  height: 80,
  color: Colors.green,
  child: Center(
    child: Text(
      'This is a Container',
      style: TextStyle(color: Colors.white, fontSize: 18),
    ),
  ),
),
SizedBox(height: 10),

// Row Widget
Row(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
  children: [
    Icon(Icons.home, size: 40, color: Colors.red),
    Icon(Icons.star, size: 40, color: Colors.yellow),
    Icon(Icons.person, size: 40, color: Colors.blue),
  ],
),
SizedBox(height: 10),

// Card Widget
Card(
  elevation: 4,
  child: Padding(
    padding: EdgeInsets.all(16),
    child: Text('This is a Card Widget'),
```

```
            ),
          ),
          SizedBox(height: 10),

          // Checkbox Widget
          CheckboxListTile(
            title: Text('Accept Terms'),
            value: true,
            onChanged: (val) {},
          ),

          // Switch Widget
          SwitchListTile(
            title: Text('Enable Notifications'),
            value: false,
            onChanged: (val) {},
          ),
        ],
      ),
    ),
  );
}
}
```

EXP 3: Create an Interactive Form Using Form Widget

File: `lib/main.dart`

```dart
```

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Form Demo',
      home: FormPage(),
    );
  }
}

class FormPage extends StatefulWidget {
  @override
  _FormPageState createState() => _FormPageState();
}

class _FormPageState extends State<FormPage> {
  final _formKey = GlobalKey<FormState>();
  final _nameController = TextEditingController();
  final _emailController = TextEditingController();
  final _passwordController = TextEditingController();

  void _submitForm() {
    if (_formKey.currentState!.validate()) {
      showDialog(
        context: context,
        builder: (context) => AlertDialog(
          title: Text('Form Submitted!'),
          content: Text(
            'Name: ${_nameController.text}\nEmail: ${_emailController.text}',
          ),
          actions: [
            TextButton(
              onPressed: () => Navigator.pop(context),
              child: Text('OK'),
            ),
          ],
        ),
      );
    }
  }
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: Text('Registration Form')),
    body: Padding(
      padding: EdgeInsets.all(16),
      child: Form(
        key: _formKey,
        child: Column(
          children: [
            // Name Field
            TextFormField(
              controller: _nameController,
              decoration: InputDecoration(
                labelText: 'Full Name',
                border: OutlineInputBorder(),
                prefixIcon: Icon(Icons.person),
              ),
              validator: (value) {
                if (value == null || value.isEmpty) {
                  return 'Please enter your name';
                }
                return null;
              },
            ),
            SizedBox(height: 16),

            // Email Field
            TextFormField(
              controller: _emailController,
              decoration: InputDecoration(
                labelText: 'Email',
                border: OutlineInputBorder(),
                prefixIcon: Icon(Icons.email),
              ),
              keyboardType: TextInputType.emailAddress,
              validator: (value) {
                if (value == null || value.isEmpty) {
                  return 'Please enter your email';
                }
                if (!value.contains('@')) {
                  return 'Please enter a valid email';
                }
                return null;
              },
            ),
```

```
SizedBox(height: 16),

// Password Field
TextFormField(
  controller: _passwordController,
  decoration: InputDecoration(
    labelText: 'Password',
    border: OutlineInputBorder(),
    prefixIcon: Icon(Icons.lock),
  ),
  obscureText: true,
  validator: (value) {
    if (value == null || value.isEmpty) {
      return 'Please enter password';
    }
    if (value.length < 6) {
      return 'Password must be at least 6 characters';
    }
    return null;
  },
),
SizedBox(height: 24),

// Submit Button
SizedBox(
  width: double.infinity,
  height: 50,
  child: ElevatedButton(
    onPressed: _submitForm,
    child: Text('Submit', style: TextStyle(fontSize: 18)),
  ),
),

SizedBox(height: 10),

// Reset Button
SizedBox(
  width: double.infinity,
  height: 50,
  child: OutlinedButton(
    onPressed: () {
      _formKey.currentState!.reset();
      _nameController.clear();
      _emailController.clear();
      _passwordController.clear();
    },
    child: Text('Reset', style: TextStyle(fontSize: 18)),
```

```
                ),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

EXP 4: Design a Layout of Flutter App Using Layout Widgets

File: `lib/main.dart`

```dart
```

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Layout Demo',
      home: LayoutPage(),
    );
  }
}

class LayoutPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Layout Widgets')),
      body: SingleChildScrollView(
        child: Column(
          children: [
            // Row Layout
            Container(
              color: Colors.blue[50],
              padding: EdgeInsets.all(8),
              child: Column(
                crossAxisAlignment: CrossAxisAlignment.start,
                children: [
                  Text('ROW - Horizontal Layout',
                      style: TextStyle(fontWeight: FontWeight.bold)),
                  Row(
                    mainAxisAlignment: MainAxisAlignment.spaceAround,
                    children: [
                      _box('Box 1', Colors.red),
                      _box('Box 2', Colors.green),
                      _box('Box 3', Colors.blue),
                    ],
                  ),
                ],
              ),
            ),
            SizedBox(height: 10),
```

```
// Column Layout
Container(
  color: Colors.green[50],
  padding: EdgeInsets.all(8),
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Text('COLUMN - Vertical Layout',
          style: TextStyle(fontWeight: FontWeight.bold)),
      _box('Item 1', Colors.orange),
      SizedBox(height: 5),
      _box('Item 2', Colors.purple),
      SizedBox(height: 5),
      _box('Item 3', Colors.teal),
    ],
  ),
),
SizedBox(height: 10),

// Stack Layout
Container(
  color: Colors.orange[50],
  padding: EdgeInsets.all(8),
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Text('STACK - Overlapping Layout',
          style: TextStyle(fontWeight: FontWeight.bold)),
      SizedBox(height: 5),
      Stack(
        children: [
          Container(width: 200, height: 100, color: Colors.blue),
          Positioned(
            top: 20,
            left: 30,
            child: Container(width: 100, height: 60, color: Colors.red),
          ),
          Positioned(
            top: 35,
            left: 60,
            child: Text('Stacked!',
                style: TextStyle(color: Colors.white, fontSize: 16)),
          ),
        ],
      ),
    ],
  ),
```

```
      ),
      SizedBox(height: 10),

      // Expanded & Flexible
      Container(
        color: Colors.purple[50],
        padding: EdgeInsets.all(8),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text('EXPANDED - Flexible Layout',
                style: TextStyle(fontWeight: FontWeight.bold)),
            Row(
              children: [
                Expanded(flex: 1, child: _box('1x', Colors.pink)),
                SizedBox(width: 5),
                Expanded(flex: 2, child: _box('2x', Colors.indigo)),
                SizedBox(width: 5),
                Expanded(flex: 1, child: _box('1x', Colors.cyan)),
              ],
            ),
          ],
        ),
      ),
      SizedBox(height: 10),

      // GridView
      Container(
        color: Colors.yellow[50],
        padding: EdgeInsets.all(8),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text('GRIDVIEW - Grid Layout',
                style: TextStyle(fontWeight: FontWeight.bold)),
            GridView.count(
              crossAxisCount: 3,
              shrinkWrap: true,
              physics: NeverScrollableScrollPhysics(),
              children: List.generate(6, (index) {
                return _box('G${index + 1}', Colors.primaries[index]);
              }),
            ),
          ],
        ),
      ),
    ],
```

```
        ),
      ),
    );
  }

  Widget _box(String text, Color color) {
    return Container(
      padding: EdgeInsets.all(10),
      color: color,
      child: Text(text, style: TextStyle(color: Colors.white)),
    );
  }
}
```

EXP 5: Include Icons, Images, Charts in Flutter App

pubspec.yaml - Add dependency:

```yaml
dependencies:
  flutter:
    sdk: flutter
  fl_chart: ^0.65.0
```

Run: `flutter pub get`

File: `lib/main.dart`

```dart
```

```
import 'package:flutter/material.dart';
import 'package:fl_chart/fl_chart.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Icons Images Charts',
      home: MediaPage(),
    );
  }
}

class MediaPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Icons, Images & Charts')),
      body: SingleChildScrollView(
        padding: EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            // ICONS SECTION
            Text('Icons', style: TextStyle(fontSize: 20, fontWeight: FontWeight.bold
            SizedBox(height: 8),
            Row(
              mainAxisAlignment: MainAxisAlignment.spaceAround,
              children: [
                Column(children: [Icon(Icons.home, size: 40, color: Colors.blue), Te
                Column(children: [Icon(Icons.favorite, size: 40, color: Colors.red),
                Column(children: [Icon(Icons.share, size: 40, color: Colors.green),
                Column(children: [Icon(Icons.settings, size: 40, color: Colors.grey)
                Column(children: [Icon(Icons.camera, size: 40, color: Colors.purple)
              ],
            ),
            SizedBox(height: 20),

            // IMAGE FROM NETWORK
            Text('Network Image', style: TextStyle(fontSize: 20, fontWeight: FontWei
            SizedBox(height: 8),
            Center(
```

```
            child: ClipRRect(
              borderRadius: BorderRadius.circular(12),
              child: Image.network(
                'https://picsum.photos/300/150',
                width: 300,
                height: 150,
                fit: BoxFit.cover,
                loadingBuilder: (context, child, loadingProgress) {
                  if (loadingProgress == null) return child;
                  return CircularProgressIndicator();
                },
              ),
            ),
          ),
          SizedBox(height: 20),

          // BAR CHART
          Text('Bar Chart', style: TextStyle(fontSize: 20, fontWeight: FontWeight.
          SizedBox(height: 8),
          Container(
            height: 200,
            child: BarChart(
              BarChartData(
                barGroups: [
                  BarChartGroupData(x: 0, barRods: [BarChartRodData(toY: 8, color:
                  BarChartGroupData(x: 1, barRods: [BarChartRodData(toY: 5, color:
                  BarChartGroupData(x: 2, barRods: [BarChartRodData(toY: 9, color:
                  BarChartGroupData(x: 3, barRods: [BarChartRodData(toY: 3, color:
                  BarChartGroupData(x: 4, barRods: [BarChartRodData(toY: 7, color:
                ],
              ),
            ),
          ),
          SizedBox(height: 20),

          // LINE CHART
          Text('Line Chart', style: TextStyle(fontSize: 20, fontWeight: FontWeight
          SizedBox(height: 8),
          Container(
            height: 200,
            child: LineChart(
              LineChartData(
                lineBarsData: [
                  LineChartBarData(
                    spots: [
                      FlSpot(0, 3),
                      FlSpot(1, 5),
```

```
                    FlSpot(2, 2),
                    FlSpot(3, 8),
                    FlSpot(4, 6),
                  ],
                  color: Colors.blue,
                  isCurved: true,
                ),
              ],
            ),
          ),
        ),
      ],
    ),
  ),
);
}
}
```

EXP 6: Apply Navigation, Routing and Gestures in Flutter App

File: `lib/main.dart`

```dart
```

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Navigation Demo',
      initialRoute: '/',
      routes: {
        '/': (context) => HomePage(),
        '/second': (context) => SecondPage(),
        '/third': (context) => ThirdPage(),
      },
    );
  }
}

// HOME PAGE
class HomePage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Home Page')),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text('Home Page', style: TextStyle(fontSize: 24)),
            SizedBox(height: 20),

            // Navigate using push
            ElevatedButton(
              onPressed: () {
                Navigator.pushNamed(context, '/second');
              },
              child: Text('Go to Second Page'),
            ),
            SizedBox(height: 10),

            // Navigate with data
            ElevatedButton(
              onPressed: () {
```

```
            Navigator.push(
              context,
              MaterialPageRoute(
                builder: (context) => ThirdPage(),
              ),
            );
          },
          child: Text('Go to Third Page'),
        ),
        SizedBox(height: 20),

        // GESTURE DETECTOR
        Text('Gesture Demo:', style: TextStyle(fontSize: 18)),
        SizedBox(height: 10),
        GestureDetector(
          onTap: () {
            ScaffoldMessenger.of(context).showSnackBar(
              SnackBar(content: Text('Tapped!')),
            );
          },
          onDoubleTap: () {
            ScaffoldMessenger.of(context).showSnackBar(
              SnackBar(content: Text('Double Tapped!')),
            );
          },
          onLongPress: () {
            ScaffoldMessenger.of(context).showSnackBar(
              SnackBar(content: Text('Long Pressed!')),
            );
          },
          child: Container(
            width: 200,
            height: 80,
            color: Colors.blue,
            child: Center(
              child: Text(
                'Tap / Double Tap / Long Press',
                textAlign: TextAlign.center,
                style: TextStyle(color: Colors.white),
              ),
            ),
          ),
        ),
      ],
    ),
  ),
  // Bottom Navigation Bar
```

```
      bottomNavigationBar: BottomNavigationBar(
        items: [
          BottomNavigationBarItem(icon: Icon(Icons.home), label: 'Home'),
          BottomNavigationBarItem(icon: Icon(Icons.search), label: 'Search'),
          BottomNavigationBarItem(icon: Icon(Icons.person), label: 'Profile'),
        ],
        onTap: (index) {
          print('Tab $index tapped');
        },
      ),
    );
  }
}

// SECOND PAGE
class SecondPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Second Page')),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text('Second Page', style: TextStyle(fontSize: 24)),
            SizedBox(height: 20),
            ElevatedButton(
              onPressed: () {
                Navigator.pop(context); // Go back
              },
              child: Text('Go Back'),
            ),
          ],
        ),
      ),
    );
  }
}

// THIRD PAGE
class ThirdPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Third Page')),
      body: Center(
        child: Column(
```

```
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text('Third Page', style: TextStyle(fontSize: 24)),
            SizedBox(height: 20),
            ElevatedButton(
              onPressed: () {
                Navigator.pushNamedAndRemoveUntil(context, '/', (route) => false);
              },
              child: Text('Go to Home (Clear Stack)'),
            ),
          ],
        ),
      ),
    );
  }
}
```

EXP 7: Connect Flutter UI with Firebase Database

Steps:

1. Go to https://console.firebase.google.com
2. Create a new project
3. Add Android app → Enter package name (e.g., `com.example.myapp`)
4. Download `google-services.json` → put in `android/app/`
5. Add dependencies in `pubspec.yaml`:

```yaml
dependencies:
  flutter:
    sdk: flutter
  firebase_core: ^2.24.2
  firebase_database: ^10.4.0
```

`android/build.gradle` - Add at bottom:

```gradle
classpath 'com.google.gms:google-services:4.4.0'
```

`android/app/build.gradle` - Add at bottom:

```gradle
apply plugin: 'com.google.gms.google-services'
```

Run: `flutter pub get`

File: `lib/main.dart`

```dart
```

```dart
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:firebase_database/firebase_database.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Firebase Demo',
      home: FirebasePage(),
    );
  }
}

class FirebasePage extends StatefulWidget {
  @override
  _FirebasePageState createState() => _FirebasePageState();
}

class _FirebasePageState extends State<FirebasePage> {
  final _controller = TextEditingController();
  final _db = FirebaseDatabase.instance.ref();
  String _savedData = '';

  void _saveData() {
    _db.child('message').set(_controller.text);
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text('Data Saved!')),
    );
  }

  void _readData() {
    _db.child('message').once().then((event) {
      setState(() {
        _savedData = event.snapshot.value.toString();
      });
    });
  }

  @override
```

```
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Firebase Database')),
      body: Padding(
        padding: EdgeInsets.all(16),
        child: Column(
          children: [
            TextField(
              controller: _controller,
              decoration: InputDecoration(
                labelText: 'Enter Message',
                border: OutlineInputBorder(),
              ),
            ),
            SizedBox(height: 16),
            ElevatedButton(
              onPressed: _saveData,
              child: Text('Save to Firebase'),
            ),
            SizedBox(height: 10),
            ElevatedButton(
              onPressed: _readData,
              child: Text('Read from Firebase'),
            ),
            SizedBox(height: 20),
            Text('Data from Firebase:', style: TextStyle(fontSize: 18)),
            Text(_savedData, style: TextStyle(fontSize: 16, color: Colors.blue)),
          ],
        ),
      ),
    );
  }
}
```

EXP 8: Test and Deploy Production Ready Flutter App on Android

Steps:

1. Test the app:

```bash
flutter test
flutter analyze
```

2. Change app name in `android/app/src/main/AndroidManifest.xml`:

```xml
android:label="My App Name"
```

3. Change package name in `android/app/build.gradle`:

```gradle
applicationId "com.yourname.myapp"
```

4. Set version in `pubspec.yaml`:

```yaml
version: 1.0.0+1
```

5. Create signing keystore:

```bash
keytool -genkey -v -keystore ~/key.jks -keyalg RSA -keysize 2048 -validity 10000 -al
```

6. Create `android/key.properties`:

```properties
storePassword=yourpassword
keyPassword=yourpassword
keyAlias=key
storeFile=/home/user/key.jks
```

7. Edit `android/app/build.gradle` to add signing:

```gradle
```

```
def keystoreProperties = new Properties()
def keystorePropertiesFile = rootProject.file('key.properties')
keystoreProperties.load(new FileInputStream(keystorePropertiesFile))

android {
    signingConfigs {
        release {
            keyAlias keystoreProperties['keyAlias']
            keyPassword keystoreProperties['keyPassword']
            storeFile file(keystoreProperties['storeFile'])
            storePassword keystoreProperties['storePassword']
        }
    }
    buildTypes {
        release {
            signingConfig signingConfigs.release
        }
    }
}
```

8. Build APK:

```bash
flutter build apk --release
```

APK is at: `build/app/outputs/flutter-apk/app-release.apk`

9. Install on device:

```bash
flutter install
```

EXP 9: Responsive UI for Ecommerce Using React

File: `index.html`

```html
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>E-Commerce Shop</title>
  <style>
    * { margin: 0; padding: 0; box-sizing: border-box; }
    body { font-family: Arial, sans-serif; background: #f5f5f5; }

    /* NAVBAR */
    nav {
      background: #333;
      color: white;
      padding: 15px 20px;
      display: flex;
      justify-content: space-between;
      align-items: center;
    }
    nav .logo { font-size: 22px; font-weight: bold; }
    nav .cart { font-size: 16px; cursor: pointer; }

    /* HERO */
    .hero {
      background: linear-gradient(135deg, #667eea, #764ba2);
      color: white;
      text-align: center;
      padding: 60px 20px;
    }
    .hero h1 { font-size: 36px; margin-bottom: 10px; }
    .hero p { font-size: 18px; margin-bottom: 20px; }
    .hero button {
      background: white;
      color: #667eea;
      border: none;
      padding: 12px 30px;
      font-size: 16px;
      border-radius: 30px;
      cursor: pointer;
    }

    /* PRODUCTS GRID */
    .products {
      padding: 30px 20px;
      max-width: 1200px;
      margin: 0 auto;
```

```
  }
  .products h2 { margin-bottom: 20px; text-align: center; font-size: 28px; }
  .grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
    gap: 20px;
  }
  .card {
    background: white;
    border-radius: 10px;
    overflow: hidden;
    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
    transition: transform 0.2s;
  }
  .card:hover { transform: translateY(-5px); }
  .card img { width: 100%; height: 180px; object-fit: cover; }
  .card-body { padding: 15px; }
  .card-body h3 { font-size: 16px; margin-bottom: 8px; }
  .card-body .price { color: #e44d26; font-size: 18px; font-weight: bold; }
  .card-body button {
    width: 100%;
    margin-top: 10px;
    padding: 10px;
    background: #333;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
    font-size: 14px;
  }
  .card-body button:hover { background: #555; }

  /* FOOTER */
  footer {
    background: #333;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 30px;
  }

  /* RESPONSIVE */
  @media (max-width: 600px) {
    .hero h1 { font-size: 24px; }
    .hero p { font-size: 14px; }
  }
</style>
```

```html
</head>
<body>

  <!-- NAVBAR -->
  <nav>
    <div class="logo">🛒 ShopEasy</div>
    <div class="cart">🛍️ Cart (0)</div>
  </nav>

  <!-- HERO -->
  <div class="hero">
    <h1>Welcome to ShopEasy</h1>
    <p>Discover amazing products at great prices</p>
    <button>Shop Now</button>
  </div>

  <!-- PRODUCTS -->
  <div class="products">
    <h2>Featured Products</h2>
    <div class="grid">
      <div class="card">
        <img src="https://picsum.photos/seed/prod1/300/200" alt="Product 1"/>
        <div class="card-body">
          <h3>Wireless Headphones</h3>
          <div class="price">₹1,999</div>
          <button onclick="alert('Added to Cart!')">Add to Cart</button>
        </div>
      </div>
      <div class="card">
        <img src="https://picsum.photos/seed/prod2/300/200" alt="Product 2"/>
        <div class="card-body">
          <h3>Smart Watch</h3>
          <div class="price">₹3,499</div>
          <button onclick="alert('Added to Cart!')">Add to Cart</button>
        </div>
      </div>
      <div class="card">
        <img src="https://picsum.photos/seed/prod3/300/200" alt="Product 3"/>
        <div class="card-body">
          <h3>Laptop Bag</h3>
          <div class="price">₹899</div>
          <button onclick="alert('Added to Cart!')">Add to Cart</button>
        </div>
      </div>
      <div class="card">
        <img src="https://picsum.photos/seed/prod4/300/200" alt="Product 4"/>
        <div class="card-body">
```

```
          <h3>Bluetooth Speaker</h3>
          <div class="price">₹1,299</div>
          <button onclick="alert('Added to Cart!')">Add to Cart</button>
        </div>
      </div>
    </div>
  </div>

  <!-- FOOTER -->
  <footer>
    <p>&copy; 2024 ShopEasy. All rights reserved.</p>
  </footer>

</body>
</html>
```

EXP 10: Write Web App Manifest File for PWA (Add to Homescreen)

File: `manifest.json`

```
json
```

```json
{
  "name": "ShopEasy E-Commerce",
  "short_name": "ShopEasy",
  "description": "An e-commerce Progressive Web App",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#333333",
  "orientation": "portrait",
  "scope": "/",
  "lang": "en",
  "icons": [
    {
      "src": "icons/icon-72x72.png",
      "sizes": "72x72",
      "type": "image/png"
    },
    {
      "src": "icons/icon-96x96.png",
      "sizes": "96x96",
      "type": "image/png"
    },
    {
      "src": "icons/icon-128x128.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "icons/icon-144x144.png",
      "sizes": "144x144",
      "type": "image/png"
    },
    {
      "src": "icons/icon-192x192.png",
      "sizes": "192x192",
      "type": "image/png",
      "purpose": "any maskable"
    },
    {
      "src": "icons/icon-512x512.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ],
  "screenshots": [
    {
```

```
      "src": "screenshots/screen1.png",
      "sizes": "540x720",
      "type": "image/png"
    }
  ],
  "categories": ["shopping", "ecommerce"],
  "related_applications": []
}
```

Add to `index.html` inside `<head>`:

```html
<link rel="manifest" href="manifest.json">
<meta name="theme-color" content="#333333">
<meta name="apple-mobile-web-app-capable" content="yes">
<meta name="apple-mobile-web-app-status-bar-style" content="black">
<meta name="apple-mobile-web-app-title" content="ShopEasy">
<link rel="apple-touch-icon" href="icons/icon-192x192.png">
```

EXP 11: Code and Register a Service Worker for E-Commerce PWA

File: `service-worker.js`

```javascript
```

```
const CACHE_NAME = 'shopeasy-cache-v1';
const urlsToCache = [
  '/',
  '/index.html',
  '/style.css',
  '/app.js',
  '/manifest.json',
  '/icons/icon-192x192.png'
];

// INSTALL EVENT - Cache files
self.addEventListener('install', (event) => {
  console.log('Service Worker: Installing...');
  event.waitUntil(
    caches.open(CACHE_NAME).then((cache) => {
      console.log('Service Worker: Caching Files');
      return cache.addAll(urlsToCache);
    })
  );
  self.skipWaiting(); // Activate immediately
});

// ACTIVATE EVENT - Clean old caches
self.addEventListener('activate', (event) => {
  console.log('Service Worker: Activated');
  event.waitUntil(
    caches.keys().then((cacheNames) => {
      return Promise.all(
        cacheNames.map((cache) => {
          if (cache !== CACHE_NAME) {
            console.log('Service Worker: Clearing Old Cache -', cache);
            return caches.delete(cache);
          }
        })
      );
    })
  );
  self.clients.claim();
});
```

Add in `index.html` to Register Service Worker:

```html
```

```
<script>
  if ('serviceWorker' in navigator) {
    window.addEventListener('load', () => {
      navigator.serviceWorker.register('/service-worker.js')
        .then((registration) => {
          console.log('Service Worker registered! Scope:', registration.scope);
        })
        .catch((error) => {
          console.log('Service Worker registration failed:', error);
        });
    });
  }
</script>
```

EXP 12: Implement Service Worker Events - Fetch, Sync, Push

File: `service-worker.js` (Full version with all events)

```javascript
```

```
const CACHE_NAME = 'shopeasy-v1';
const urlsToCache = ['/', '/index.html'];

// INSTALL
self.addEventListener('install', (event) => {
  event.waitUntil(
    caches.open(CACHE_NAME).then((cache) => cache.addAll(urlsToCache))
  );
  self.skipWaiting();
});

// ACTIVATE
self.addEventListener('activate', (event) => {
  event.waitUntil(
    caches.keys().then((keys) =>
      Promise.all(keys.filter(k => k !== CACHE_NAME).map(k => caches.delete(k)))
    )
  );
  self.clients.claim();
});

// ============ FETCH EVENT ============
self.addEventListener('fetch', (event) => {
  console.log('Fetching:', event.request.url);
  event.respondWith(
    caches.match(event.request).then((cachedResponse) => {
      // Return cached if available
      if (cachedResponse) {
        return cachedResponse;
      }
      // Otherwise fetch from network
      return fetch(event.request).then((networkResponse) => {
        // Cache the new response
        return caches.open(CACHE_NAME).then((cache) => {
          cache.put(event.request, networkResponse.clone());
          return networkResponse;
        });
      }).catch(() => {
        // Offline fallback
        return new Response('<h1>Offline - No Internet</h1>', {
          headers: { 'Content-Type': 'text/html' }
        });
      });
    })
  );
});
```

```
// ============ SYNC EVENT ============
self.addEventListener('sync', (event) => {
  console.log('Background Sync:', event.tag);
  if (event.tag === 'sync-cart') {
    event.waitUntil(syncCartData());
  }
  if (event.tag === 'sync-orders') {
    event.waitUntil(syncOrders());
  }
});

async function syncCartData() {
  console.log('Syncing cart data with server...');
  // In real app: get data from IndexedDB, send to server
  return fetch('/api/sync-cart', {
    method: 'POST',
    body: JSON.stringify({ synced: true }),
    headers: { 'Content-Type': 'application/json' }
  });
}

async function syncOrders() {
  console.log('Syncing orders...');
  return fetch('/api/sync-orders', { method: 'POST' });
}

// ============ PUSH EVENT ============
self.addEventListener('push', (event) => {
  console.log('Push notification received');

  let data = { title: 'ShopEasy', body: 'New deal available!', icon: '/icons/icon-19

  if (event.data) {
    data = event.data.json();
  }

  const options = {
    body: data.body,
    icon: data.icon || '/icons/icon-192x192.png',
    badge: '/icons/icon-72x72.png',
    vibrate: [100, 50, 100],
    data: { url: data.url || '/' },
    actions: [
      { action: 'view', title: 'View Deal' },
      { action: 'dismiss', title: 'Dismiss' }
    ]
```

```
  };

  event.waitUntil(
    self.registration.showNotification(data.title, options)
  );
});

// NOTIFICATION CLICK EVENT
self.addEventListener('notificationclick', (event) => {
  event.notification.close();
  if (event.action === 'view') {
    event.waitUntil(clients.openWindow(event.notification.data.url));
  }
});
```

Register Sync and Push in `app.js`:

```javascript
// Register Background Sync
async function registerSync() {
  if ('serviceWorker' in navigator && 'SyncManager' in window) {
    const sw = await navigator.serviceWorker.ready;
    await sw.sync.register('sync-cart');
    console.log('Background Sync registered!');
  }
}

// Register Push Notifications
async function registerPush() {
  if ('serviceWorker' in navigator && 'PushManager' in window) {
    const sw = await navigator.serviceWorker.ready;
    const permission = await Notification.requestPermission();
    if (permission === 'granted') {
      const subscription = await sw.pushManager.subscribe({
        userVisibleOnly: true,
        applicationServerKey: 'YOUR_VAPID_PUBLIC_KEY' // Replace with real key
      });
      console.log('Push Subscription:', JSON.stringify(subscription));
    }
  }
}

registerSync();
registerPush();
```

EXP 13: Deploy Ecommerce PWA to GitHub Pages

Steps:

1. Create a GitHub account at https://github.com
2. Create a new repository (e.g., `shopeasy-pwa`)
3. Install Git: https://git-scm.com

```bash
# Initialize git in your project folder
git init

# Add all files
git add .

# Commit
git commit -m "Initial PWA commit"

# Add remote repository (replace with YOUR github URL)
git remote add origin https://github.com/yourusername/shopeasy-pwa.git

# Push to GitHub
git push -u origin main
```

4. On GitHub → Go to your repo
5. Click **Settings** → **Pages**
6. Under **Source**: Select `main` branch → folder `/(root)`
7. Click **Save**
8. Your site will be live at: `https://yourusername.github.io/shopeasy-pwa`

Important for PWA on GitHub Pages:

- Update `start_url` in manifest.json to `/shopeasy-pwa/`
- Update service worker cache paths to include `/shopeasy-pwa/`
- Make sure all file paths are relative

EXP 14: Use Google Lighthouse PWA Analysis Tool

Steps:

Method 1 - Chrome DevTools (Easiest):

1. Open your PWA in Chrome browser
2. Press `F12` or Right-click → **Inspect**
3. Click the **Lighthouse** tab
4. Select categories: **Performance**, **PWA**, **Accessibility**, **SEO**
5. Select device: **Mobile** or **Desktop**
6. Click **Analyze page load**
7. Wait for analysis to complete
8. View scores (aim for green: 90+)

Method 2 - Command Line:

```bash
# Install Lighthouse
npm install -g lighthouse

# Run analysis
lighthouse https://yourusername.github.io/shopeasy-pwa --view

# With specific output
lighthouse https://yourusername.github.io/shopeasy-pwa --output html --output-path r
```

What Lighthouse checks for PWA:

- ✅ Has a valid manifest.json
- ✅ Service Worker is registered
- ✅ Works offline
- ✅ HTTPS is used
- ✅ Icons are correct size
- ✅ Has meta viewport tag
- ✅ Has theme-color meta tag

How to improve scores:

- Add proper icons (all sizes)

- Enable HTTPS
- Add offline fallback page
- Compress images
- Minimize CSS/JS

EXP 15: Deploy Ecommerce PWA Using SSL-Enabled Static Hosting

Option A: Netlify (Easiest - Free SSL)

Steps:

1. Go to https://netlify.com → Sign up free
2. Click **Add new site** → **Deploy manually**
3. Drag and drop your project folder
4. Netlify auto-deploys with free HTTPS/SSL!
5. Your site gets URL like: `https://random-name.netlify.app`

Or using Netlify CLI:

```bash
# Install Netlify CLI
npm install -g netlify-cli

# Login
netlify login

# Deploy
netlify deploy --prod --dir=.
```

Option B: Vercel (Also Free SSL)

```bash
# Install Vercel CLI
npm install -g vercel

# Deploy
vercel --prod
```

Option C: Firebase Hosting (Free SSL)

```bash
```

```
# Install Firebase CLI
npm install -g firebase-tools

# Login
firebase login

# Initialize project
firebase init hosting

# Deploy
firebase deploy
```

Verify SSL is Working:

1. Open your site URL (starts with `https://`)
2. Click the 🔒 lock icon in browser address bar
3. Should show "Connection is secure"

`netlify.toml` config file (for better PWA support):

```toml
[[headers]]
  for = "/service-worker.js"
  [headers.values]
    Cache-Control = "no-cache"

[[headers]]
  for = "/*"
  [headers.values]
    X-Frame-Options = "DENY"
    X-XSS-Protection = "1; mode=block"
```

QUICK REFERENCE

Exp	Topic	Type
1	Flutter Setup	Steps
2	Common Widgets	Flutter Code
3	Form Widget	Flutter Code
4	Layout Widgets	Flutter Code

Exp	Topic	Type
5	Icons/Images/Charts	Flutter Code
6	Navigation/Routing/Gestures	Flutter Code
7	Firebase Integration	Flutter Code
8	Build & Deploy Android	Steps
9	Responsive E-commerce UI	HTML/CSS Code
10	Web App Manifest (PWA)	JSON Code
11	Service Worker Register	JS Code
12	Fetch/Sync/Push Events	JS Code
13	Deploy to GitHub Pages	Steps
14	Lighthouse Analysis	Steps
15	SSL Static Hosting	Steps